Experiment - 8

Data Visualization I

1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see if we can find any patterns in the data.
2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram.

In [1]:

```
pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\hp\anaconda3\lib\site-packages (0.
9.0)
Requirement already satisfied: matplotlib>=1.4.3 in c:\users\hp\anaconda3\lib\site-pac
kages (from seaborn) (3.1.1)
Requirement already satisfied: numpy>=1.9.3 in c:\users\hp\anaconda3\lib\site-packages
(from seaborn) (1.16.5)
Requirement already satisfied: scipy>=0.14.0 in c:\users\hp\anaconda3\lib\site-package
s (from seaborn) (1.3.1)
Requirement already satisfied: pandas>=0.15.2 in c:\users\hp\anaconda3\lib\site-packag
es (from seaborn) (0.25.1)
Requirement already satisfied: cycler>=0.10 in c:\users\hp\anaconda3\lib\site-packages
(from matplotlib>=1.4.3->seaborn) (0.10.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\hp\anaconda3\lib\site-pac
kages (from matplotlib>=1.4.3->seaborn) (1.1.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!=2.1.6,>=2.0.1 in c:\users\hp
\anaconda3\lib\site-packages (from matplotlib>=1.4.3->seaborn) (2.4.2)
Requirement already satisfied: python-dateutil>=2.1 in c:\users\hp\anaconda3\lib\site-
packages (from matplotlib>=1.4.3->seaborn) (2.8.0)
Requirement already satisfied: pytz>=2017.2 in c:\users\hp\anaconda3\lib\site-packages
(from pandas>=0.15.2->seaborn) (2019.3)
Requirement already satisfied: six in c:\users\hp\anaconda3\lib\site-packages (from cy
cler>=0.10->matplotlib>=1.4.3->seaborn) (1.12.0)
Requirement already satisfied: setuptools in c:\users\hp\anaconda3\lib\site-packages
(from kiwisolver>=1.0.1->matplotlib>=1.4.3->seaborn) (41.4.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:

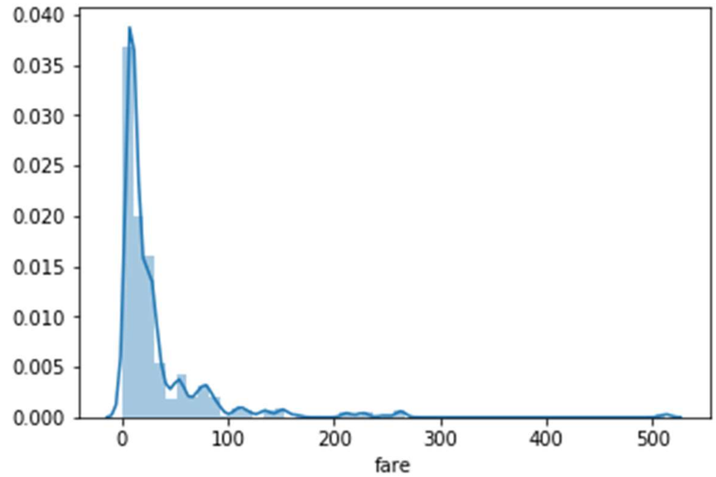
```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

dataset = sns.load_dataset('titanic')

dataset.head()
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	emb
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Sou
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	C
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Sou
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Sou
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Sou

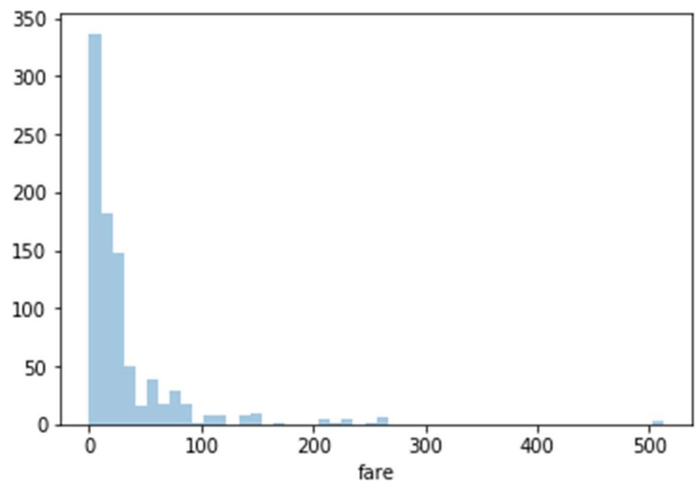
In [3]:
```
sns.distplot(dataset['fare'])
```

Out[3]: <matplotlib.axes._subplots.AxesSubplot at 0x2367329efc8>



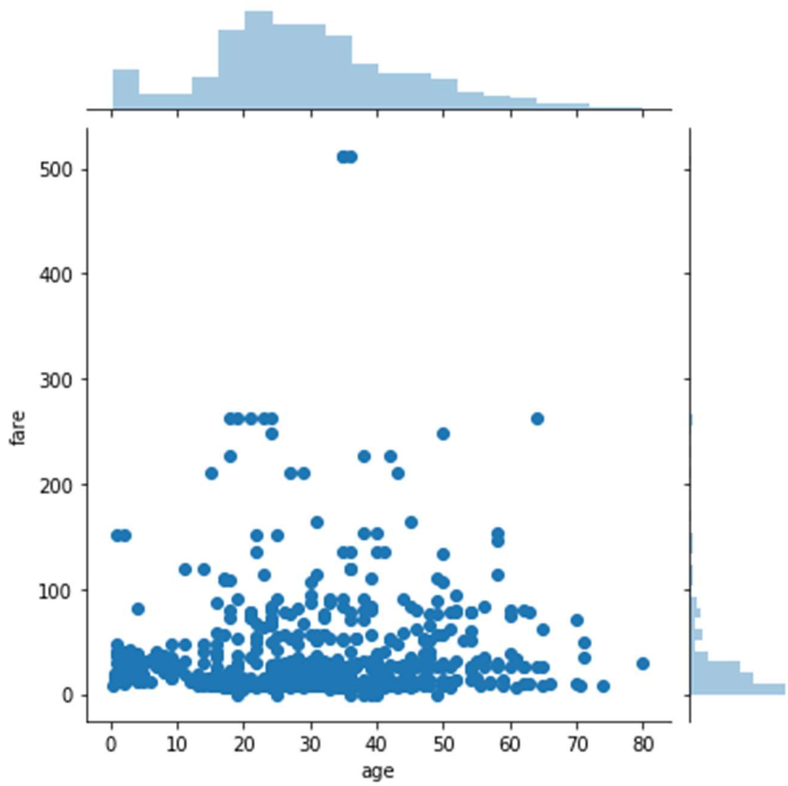
In [4]:
```
sns.distplot(dataset['fare'], kde=False)
```

Out[4]: <matplotlib.axes._subplots.AxesSubplot at 0x236736415c8>



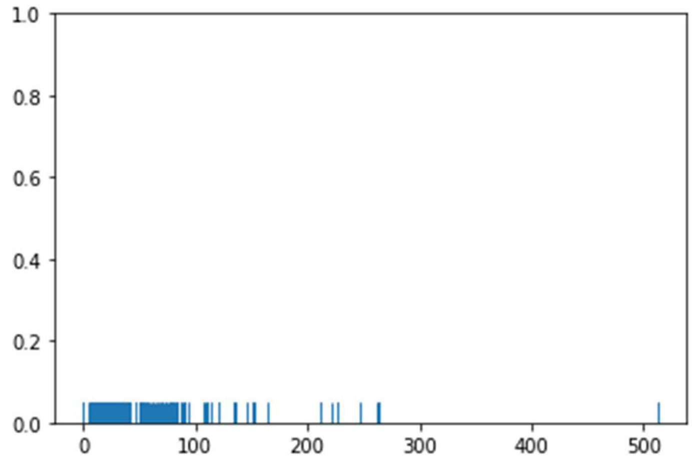
In [5]:
```
sns.jointplot(x='age', y='fare', data=dataset)
```

Out[5]: <seaborn.axisgrid.JointGrid at 0x23673741888>



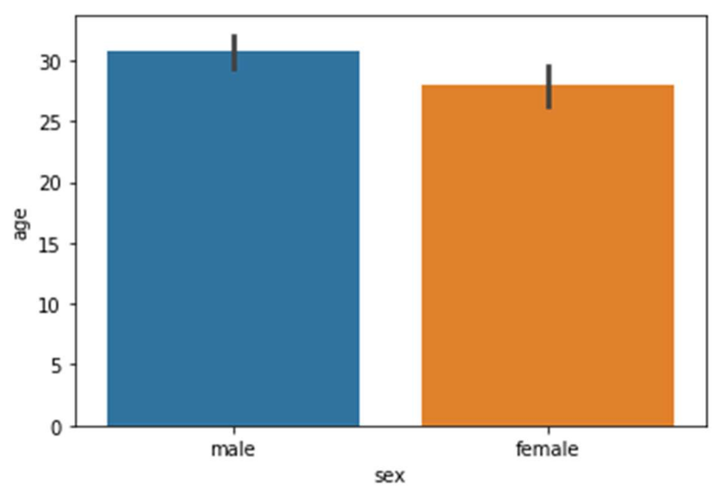
In [6]:
```
sns.rugplot(dataset['fare'])
```

Out[6]: <matplotlib.axes._subplots.AxesSubplot at 0x23673972c08>



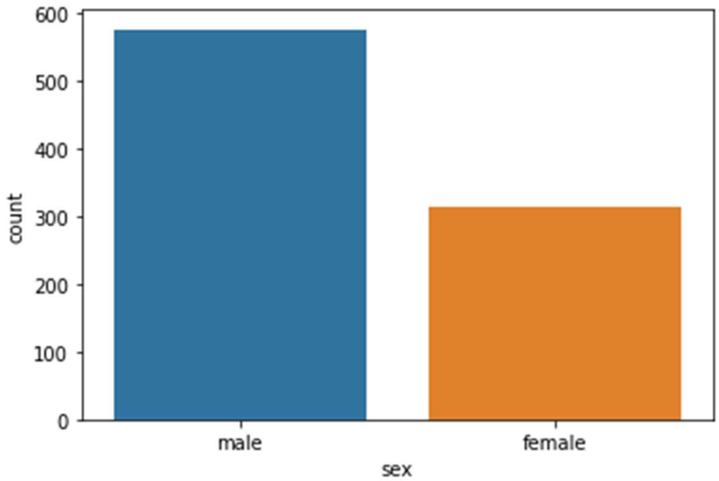
In [7]:
```
sns.barplot(x='sex', y='age', data=dataset)
```

Out[7]: <matplotlib.axes._subplots.AxesSubplot at 0x23673a2f648>



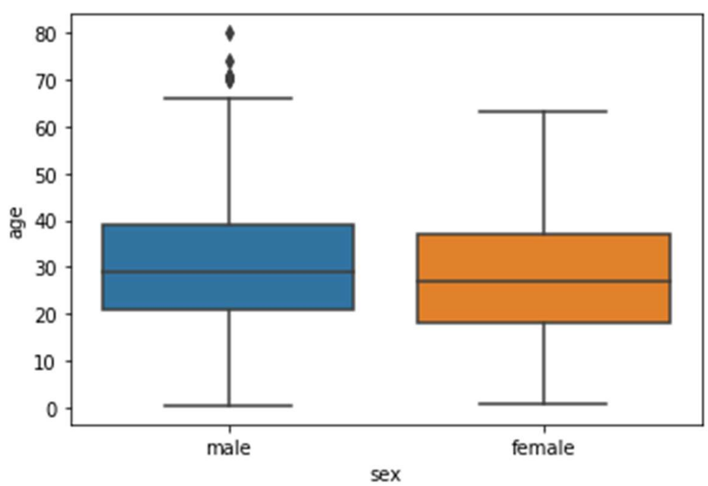
In [8]:
```python
sns.countplot(x='sex', data=dataset)
```

Out[8]: <matplotlib.axes._subplots.AxesSubplot at 0x2366d724a48>



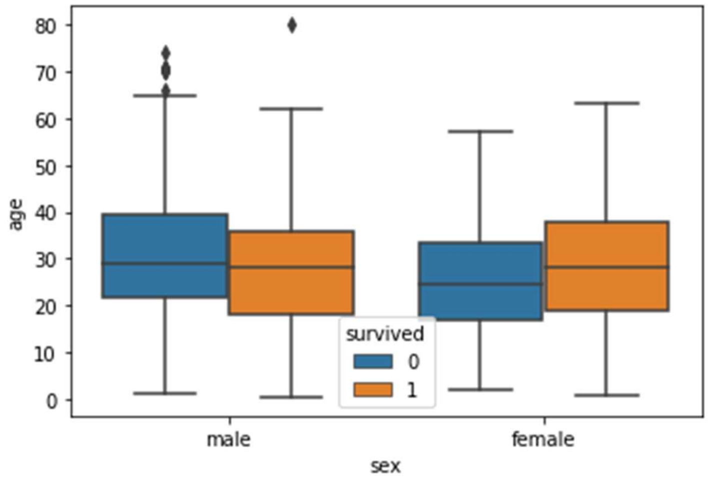
In [9]:
```python
sns.boxplot(x='sex', y='age', data=dataset)
```

Out[9]: <matplotlib.axes._subplots.AxesSubplot at 0x23673afc208>



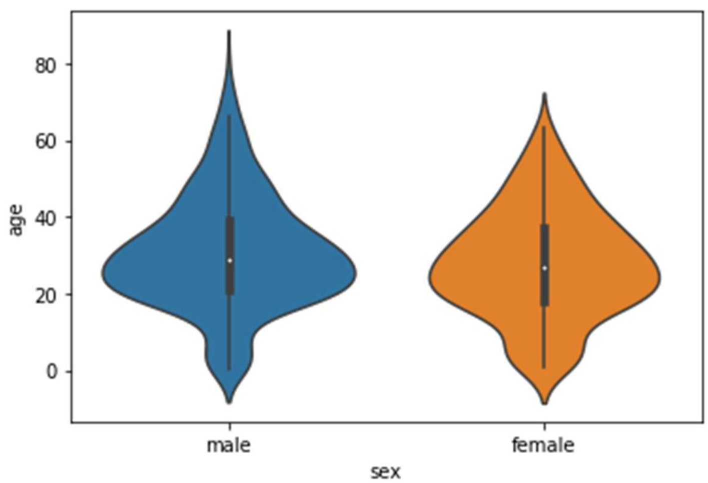
In [10]:
```
sns.boxplot(x='sex', y='age', data=dataset, hue="survived")
```

Out[10]: <matplotlib.axes._subplots.AxesSubplot at 0x23673b80508>



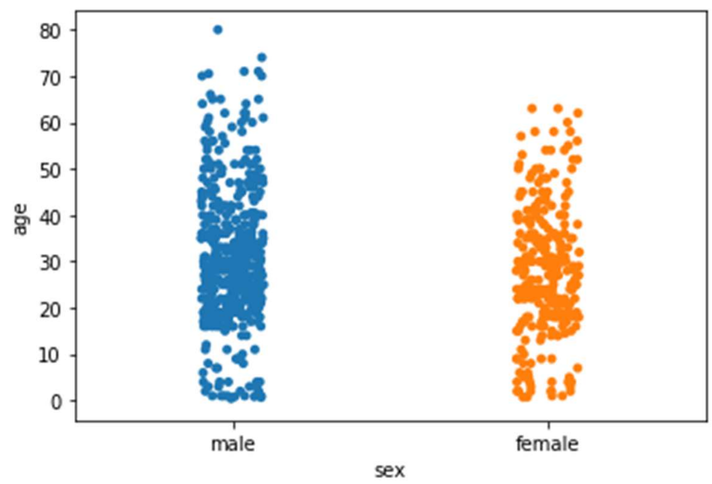
In [11]:
```
sns.violinplot(x='sex', y='age', data=dataset)
```

Out[11]: <matplotlib.axes._subplots.AxesSubplot at 0x23673c46e48>



In [12]:
```
sns.stripplot(x='sex', y='age', data=dataset)
```

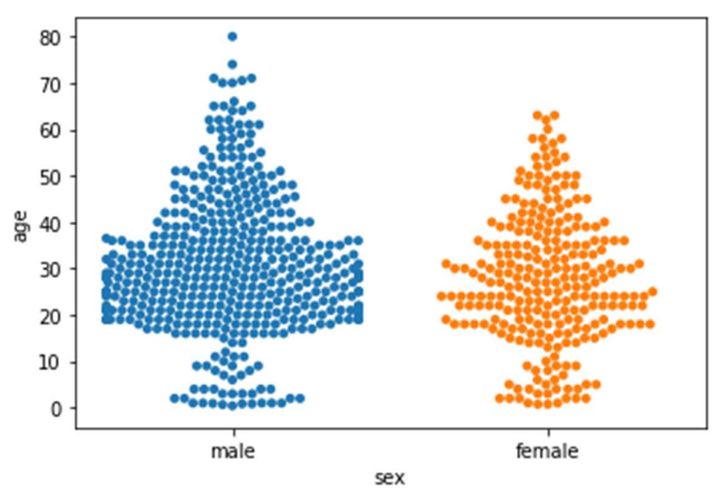
Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x23673ca1888>



In [13]:
```
sns.swarmplot(x='sex', y='age', data=dataset)
```

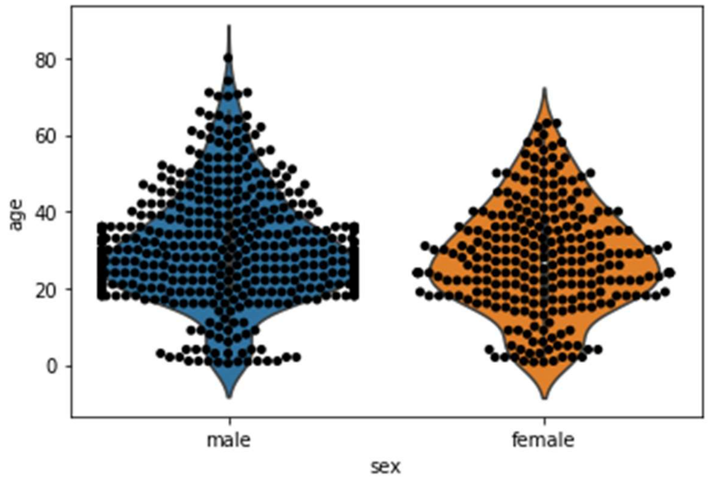
```
C:\Users\HP\Anaconda3\lib\site-packages\seaborn\categorical.py:1324: RuntimeWarning: i
nvalid value encountered in less
  off_low = points < low_gutter
C:\Users\HP\Anaconda3\lib\site-packages\seaborn\categorical.py:1328: RuntimeWarning: i
nvalid value encountered in greater
  off_high = points > high_gutter
```

Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x23673d17488>

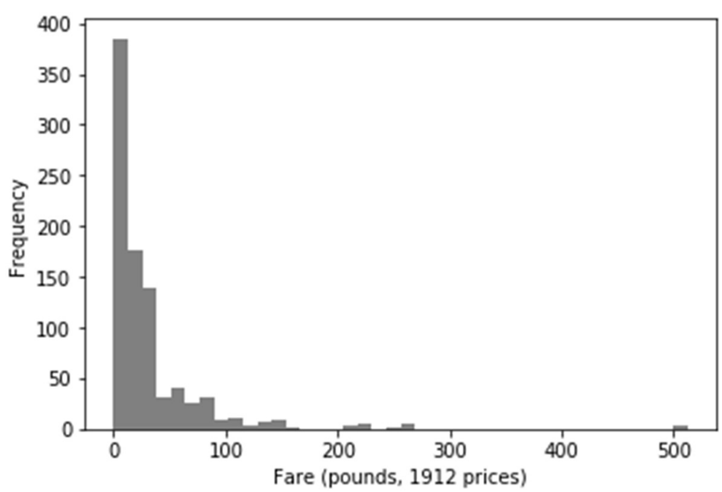


In [14]:
```
sns.violinplot(x='sex', y='age', data=dataset)
sns.swarmplot(x='sex', y='age', data=dataset, color='black')
```

Out[14]: <matplotlib.axes._subplots.AxesSubplot at 0x23673dc5e48>



In [15]:
```
#Expt. No. 8 Part-2
# histogram of fare
titanic_hist = dataset.fare.plot.hist(bins = 40, color = 'grey')
plt.xlabel('Fare (pounds, 1912 prices)')

plt.show(titanic_hist)
```



In [16]:
```
sns.distplot(dataset['fare'])
```

Out[16]: <matplotlib.axes._subplots.AxesSubplot at 0x23673c28d48>